

			Total Pages: 4
Scheme of Valuation/Answer Key			
<i>Scheme of evaluation (marks in brackets) and answers of problems/key</i>			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
THIRD SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2020			
Course Code: BMT205			
Course Name: ANALOG ELECTRONICS			
Max. Marks: 100			Duration: 3 Hours
PART A			
		Answer all questions. Each question carries 3 marks	Marks
1	1. Transistor beta 2. V_{BE} 3. I_{CO} (reverse saturation current) (Each point carries 1 mark)		(3)
2	h_{11} (h_i) h_{12} (h_r) h_{21} (h_f) h_{22} (h_o) (Writing all four parameters 2marks, Definition of each parameter 1 marks)		(3)
3	1. The human ear does not respond in a linear fashion to changes in source power level. 2. The ear, therefore, responds in a logarithmic fashion to changes in audio power levels. 3. $G_{dB} = 10 \log_{10} \frac{P_2}{P_1}$ dB (Each point carries 1 mark)		(3)
4	1. high input impedance 2. fast switching speeds 3. lower operating power levels (Each point carries 1 mark)		(3)
5	1. CMRR- Common Mode Rejection Ratio $CMRR = \frac{A_d}{A_c}$ 2. A_d - Differential gain of the amplifier and A_c -Common mode gain of the amplifier. 3. Larger the value of CMRR, better is the circuit operation (Each point carries 1 mark, the last point can be another point relevant to CMRR)		(3)
6	Neat sketch of zero crossing detector using op-amp, with proper labelling (3 marks)		(3)
7	Out of Syllabus question, 3 marks shall be awarded to everyone		(3)

8	1. It is used to measure the peak of non-sinusoidal signals. 2. It is used in the analysis of spectral and mass spectrometer. 3. Peak detector finds its application in destructive testing. 4. It is used for instrumentation measurement, mostly in amplitude modulated wave communication. 5. It widely finds applications in sound measuring instruments. (Any 3 points, 1 mark for each point)	(3)
9	1. <i>Gain margin</i> (GM) is defined as the negative of the value βA in decibels at the frequency at which the phase angle is 180°. 2. <i>Phase margin</i> (PM) is defined as the angle of 180° minus the magnitude of the angle at which the value βA is unity (0 dB). (Each point carries 1.5 marks)	(3)
10	Neat circuit diagram with labelling all the components (3 marks)	(3)
PART B		
<i>Answer any one full question from each module. Each question carries 14 marks</i>		
Module 1		
11	a. List any two/three applications of CE and CB configuration. No comparison required. (Total 5 applications. Each application carries 1 marks) b. Neat circuit diagram of voltage divider configuration (3 marks) Expression for stability factor with respect to β, V_{BE}, I_{CO} (Total 6 marks, 2 marks for each expression)	(5) (9)
12	1. Definition of Q-point (2 marks) 2. Sketching the output characteristics of CE configuration, marking the Q points, operating regions and explain the selection of Q-point (6marks) 3. Explanation on the affect of Q-point on output volrage swing (3 marks) 4. Need for the stability of operating point (3 marks)	(14)
Module 2		
13	1. Sketching the diagram indicating the construction of Depletion type MOSFET and Enhancement type MOSFET (Total 6 marks, 3 marks for each diagram) 2. Explanation and absence of channel as a constructed component of device (2 marks) 3. Sketching the transfer and drain characteristics of Depletion type MOSFET and Enhancement type MOSFET and explanation (Total 6 marks, 3 marks for each diagram)	(14)

